

Safety

PIA Safety e-Magazine

4

JUL - 2017



Greetings!

"Charity begins at home" is the slogan of a lot of welfare organizations and I shall say it is one of the most useful ones when it comes to safety.

We all including me, are not perfect and we know this but, at least we can start our journey towards excellence (not perfection) one step at a time.

As Martin Luther King had used the phrase "I have a dream" in his historical speech that changed the ideology of more than half of the people of Americas, I shall say that we all must start with same individual alacrity and substantiate it to ourselves that we BELIEVE in being better, we BELIEVE in improving individually on everything including safety involved in our jobs.

No society or organization is perfect or better word it not even near perfect but, people who are part of these behemoth groups are the ones whom make it succeed or to degrade despite whatever the limitations may be.

So, let's start paving our way to safety, to betterment, to excellence and to success.. Let's bring the change we want to see.



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CHIEF SAFETY OFFICER NOTE



Dear Readers,

Once again we have come up with another issue of safety magazine, having some articles very well written by local and foreign authors and these have been apparently very well researched and presented to you for useful knowledge.

As a safety official of the airline, me and my team at Safety & QA are of the view that it is very important to disseminate information not only to employees of our airline but, also anyone and everyone who might be interested in getting hold of a copy of the magazine and hence, for the same reason we have kept the magazine free of cost and paperless which also keeps it environment friendly.

With the experience that I have earned in the aviation industry it is pertinent to mention that we shall learn from our mistakes and move on and move forward with all the zeal and allegiance to our purpose but, at the same time it is also important for us to keep any personal grudges aside and extirpate friction in order to avoid harm to the main cause of increasing productivity or enhancing safer operations.

Best of luck!

Captain Naveed A. Aziz
Chief Safety Officer





Change on Air Safety

By Ian Fyfe-Gree BSc MSc CMgr FCMI FInstLM, post-graduate student at the University of Portsmouth, England | www.ifgairsafety.co.uk | ian.fyfe-green@myport.ac.uk

Introduction:

September 11th 2001 was not the first time in aviation history that an act of terrorism caused the deaths of hundreds of innocent passengers. The Lockerbie disaster in December 1988 was not the first time that a PanAm jumbo jet had crashed killing every passenger on board. In March 1977 at Los Rodeos airport in Tenerife two heavily laden airliners collided on the ground killing 583 people as a result of a series of events, each of which potentially insignificant, but when combined in sequence formed a catastrophic chain of errors. In October 2001 a Scandinavian Airlines MD87 collided with a Cessna 525 on the runway at Milan Linate killing 114 passengers, crew and ground staff and in 2002 a mid-air collision over Uberlingen between a Boeing 757 and a Tupolev TU164M killed 71 more. Lessons were clearly not being learnt and operational safety was being compromised. Is failure inevitability in a complex and hazardous industry?

Literature Review:

Busby and Bennett identify a number of determinants that were causatory to the Uberlingen mid-air collision, categorising them as 'functional' or 'behavioural' (2007). Within the functional section, they list interactions between systemic fault and structural vulnerabilities, whilst in behaviours they cite a lack of professional rigour and redundancy, with evidence directed towards management failures. Similarly, with the runway collision at Linate, they list evidence that exactly correlates with the categories found from Uberlingen, with structural, managerial and individual behavioural traits or faults. The prevention of incidents relies on the continuous monitoring of safety as a dynamic phenomenon (Mearns, Flin, & O'Connor, 2001). To achieve the right balance of attitude and belief, the perception of risk must be shared between employees and management. This shared responsibility shows how conflict and misunderstanding can be linked to the concept of Crew Resource Management (CRM) in human factors training.

Labib suggests that near-misses can be presented as either failures or successes dependent on the interpretation applied to them, including the interpretation of the word failure itself (2014). He also suggests that whereas a high impact low frequency event can be disastrous, it is repeated high frequency events that indicate a failing Safety Culture.

Kontogiannis and Malakis (2009, p. 693) argue that "total elimination of human error may be difficult to achieve" and that the cognitive strategies associated with the detection of error, alongside a safety culture, form the basis for training in error management.

When learning from failures, it is noted by Olive, O'Connor & Mannan (2006) that immediately after an incident the focus will be to change the safety climate within an organization. However, if the safety culture is not robust enough to support the climate then any changes could prove futile. They offer an example of this as being the "duality of the Challenger and Columbia disasters experienced by NASA" (p. 133). Labib describes in the analysis from the two space shuttle accident reports, the flawed decision making processes that were found in both, including a lack of foresight and hindsight, with no assimilation of the lessons identified in the earlier accident. It was concluded in the analysis that both reports were caused by human error and extremely similar, showing that learning had not occurred. With regards to the safety climate, the point is made that for an organization whose values are founded on curiosity of the universe NASA seemed unable to apply this same inquisitiveness and scientific rigour to itself (2014). The same similarities can also be found when looking at other organizational accidents such as BP with Texas City and Deepwater Horizon. Labib (2014, p. 77) lists lessons learned in the latter which include a lack of effective communication, poor safety management and "insufficient organizational cultural change". Inconsequential errors need to be captured and reported in order to prevent a future version of events where the outcome could be very different. It is Labib, in describing learning from error, that mentions Black Swan events with Taleb (2007) explaining the principal of a Black Swan as being an outlier when considered against the field of normal expectation, usually because there was no indication beforehand of its possibility. He also states that Black Swan events have enormous impact and that human nature will often try to rationalize the incident with hindsight with no

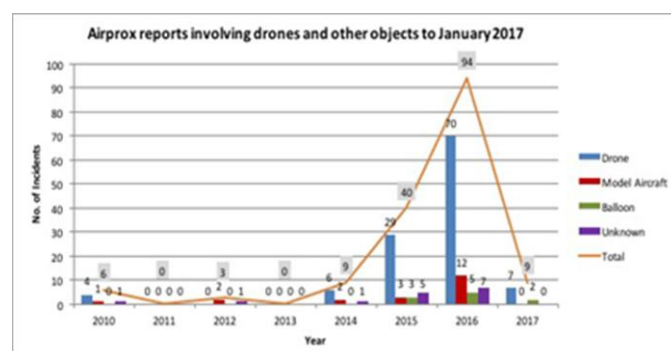
assimilation of the lessons identified in the earlier accident. It was concluded in the analysis that both reports were caused by human error and extremely similar, showing that learning had not occurred. With regards to the safety climate, the point is made that for an organization whose values are founded on curiosity of the universe NASA seemed unable to apply this same inquisitiveness and scientific rigour to itself (2014). The same similarities can also be found when looking at other organizational accidents such as BP with Texas City and deep water horizon. Labib (2014, p. 77) lists lessons learned in the latter which include a lack of effective communication, poor safety management and "insufficient organizational cultural change". Inconsequential errors need to be captured and reported in order to prevent a future version of events where the outcome could be very different. It is Labib, in describing learning from error, that mentions Black Swan events with Taleb (2007) explaining the principal of a Black Swan as being an outlier when considered against the field of normal expectation, usually because there was no indication beforehand of its possibility. He also states that Black Swan events have enormous impact and that human nature will often try to rationalize the incident with hindsight and therefore attempt to make it predictable. Labib uses this term in relation to High Reliability Organizations (HRO) and specifically when describing the events at Fukushima (2015), which Taleb describes as extremist in that Black Swans occur infrequently, but have enormous consequence on history. Taleb (2007, p.xxi) questions what the lessons learnt from 9/11 were, saying that this should ideally be "that some events, owing to their dynamics, stand largely outside the realm of the predictable". He opines that instead of this the lessons learned are entirely related to the precise avoidance of terrorism with relation to tall buildings. He goes on to say that "we do not spontaneously learn that we don't learn that we don't learn."

The aviation industry has for many years been subject to a steady change in its regulations. Transformational changes of regulations can involve completely new strategies and involves organizations doing things differently rather than just better. Kotter's model of change is considered by Flouris & Kucuk Yilmaz to emphasise that an organization confronted with transformational change must consider a "reworking of the organization's mission and strategy, its leadership and its organizational culture" (2009, p. 12). They conclude by stating that "holistic change management integration" (p.14) is required in order to introduce change without increasing risk. For any system to be considered truly functional, it must look from the perspective of overall assessment of all elements including the interactions between the human and technical systems (Cacciabue & Vella, 2010).

Furthermore, it is suggested that to do this there are two paramount points that must be compared, how risk is evaluated and how this evaluation is audited.

It is these different perspectives on risk and safety between managers and workers that underpins the theoretical framework analysis in the Bakx and Richardson paper looking at risk assessments in the Royal Netherlands Air Force (2012). They believe that significant factors in risk assessments can perceptually vary dependent on hierarchical level, citing the difference between an operator who physically bears the operating risk vice the level of risk that management deem acceptable. The acceptance of this risk therefore depends on the level to which the risk-bearers trust their management. When considering risk severity versus risk frequency (the usual method by which a risk assessment is conducted), Bakx and Richardson found that the perception of severity remained consistent between worker and manager but the frequency with which a risk event could occur was deemed "consistently slightly higher at the operator level than at the headquarter level" (2012, p. 601). Woods, Dekker, Cook, Johannsen & Sarter (2010, p.xviii) describe "an almost irresistible notion that we are custodians of already safe systems that need protection from unreliable, erratic human beings" with a list of human flaws such as perception, tiredness and irritability. They open their preface (p.xv) with "Human error is a very elusive concept"; the broad array of interpretations that this concept has leads to organizations thinking that they can achieve safety if they seek out and eradicate human error. They also comment that attribution of an error to someone is "fundamentally a social and psychological process, not an objective, technical one." A different viewpoint is offered by Dillon- Merrill, Tinsley, & Cronin, (2012) where they conclude that capturing near-miss data makes people believe more in the potential positive outcomes from the resilient misses thereby leading them to make riskier decisions and complacency.

According to data from the UK Airprox Board (2017), the number of Airprox reports filed in UK airspace between manned and unmanned aviation, rose exponentially in 2016. This can be seen in the figure below, which shows the data up to January 2017.



In their magazine (UK Airprox Board, 2016) the opening article covers the increasing prevalence of drones, and in it comment is made that “not only are the numbers of incidents increasing but the close shaves seem to be getting even closer” as well as the lack of understanding of the current regulations.

:: Research ::

From the review of the literature, the following hypotheses were established:

1. There is a temptation to rationalize every incident with hindsight bias and not acknowledge that some events are ‘Black Swans’.
2. Humans make safety through their efforts and expertise. This implies that new regulation can be safely introduced through human endeavour and training.
3. Errors occur, only if left unmanaged do they increase risk.
4. Eradicating Human Error will achieve safety of operations.
5. There is rarely a single cause of a failure event, more commonly a chain of events.
6. Operators frequently deal with information overload; new regulations can exacerbate this.
7. The capture of minor inconsequential errors is necessary to prevent a future version of events with a different (worse) outcome.
8. Using hindsight bias, all negative events are labelled with a cause.
9. Operators perceive the frequency of risk possibility to be higher than management does.
10. There is not enough regulation on the use of drones and more is needed urgently.

These hypotheses are being tested with a web based survey questionnaire aimed at members of the four worlds of the aviation industry, namely Aircrew, Engineers, Air Traffic and other support staff. To date, the survey (<http://www.ifgairsafety.co.uk/how-can-you-help/>) has been completed by 350 people from 55 different countries. Initial analysis of the results so far, shows the following:

1. The temptation to rationalize every incident is proven by the result that 63% of respondents believe that events are never random and 25% support this by saying that randomness is not very applicable to aviation.
2. Overwhelmingly, 94% of respondents believe that safety can be enhanced or made through their own endeavours.
3. 91% believe that errors are an inevitable part of day-to-day business in the aviation industry and 65% of all respondents feel that it is how errors are managed and dealt with that determines whether or not an error becomes an incident.
4. Only 20% of respondents believe that aviation operations are intrinsically safe, with 3% thinking it possible to completely remove human error.

5. Proof that it is common for there to be a chain of events that causes an incident is demonstrated by 90% of respondents saying that they had witnessed a time when a barrier had failed but another had prevented the incident occurring.

6. Whilst 77% of people had at some point in their career felt overloaded by the amount of change being introduced, 59% of these confirmed that it was because of the amount of regulatory change. Interestingly, 5% felt that the change could not have been introduced any better than it was.

7. 49% of respondents thought that it was always important to identify all errors, no matter how trivial, in order to prevent future accidents. This was supported by 38% choosing “most of the time” as their answer. Encouragingly, 64% stated that most or all the time a well-run Safety Management System captured every error and analysed them appropriately, whilst a further 26% agreed that this happens sometimes. Disappointingly, 4% felt that this never happened.

8. This hypothesis is somewhat disproven by the results of the survey, because even though 90% of respondents report that hindsight bias has an impact on safety investigations, only 13% state that a single cause is determined.

9. Most respondents felt that their management perceived risk possibilities as either the same as them or higher (62%), therefore disproving the hypothesis. Encouragingly though, 80% stated that they felt that they had a Just Culture within their organisation.

10. With regards to the amount of drone regulation in existence, only 1% felt that there was too much regulation, with 41% stating that extant regulations need some development and 43% saying that more regulation was needed urgently.

Another theme that has come out of the survey so far is that of endemic fatigue throughout the industry. This is not just tiredness, or even longstanding overwork, but many people are reporting change-fatigue, which seems prevalent across the global industry. Fatigue formed part of the article Safety & Security in the September 2016 edition of this magazine, saying that “It is a human problem that we tend to underestimate our level of fatigue and overestimate our ability to cope with it”.



:: Conclusion ::

How many times have you heard someone say, “not another change in regulations”? So how do we in the aviation industry manage changes better in the future? That is what I hope to find out in my thesis, and perhaps propose a better framework for the future introduction of change. The biggest changes that lay ahead are how we deal with the following two questions: “What will aviation regulation look like after the UK leaves the EU”, and “what do we do about drones”? The first is entirely our problem; the latter I suggest is of more global concern.

Do other countries share our concerns for the safety of manned aviation? Should drones be subject to mandatory registration? Should they carry high-intensity lighting, or perhaps electronic conspicuity? The counter-argument is that collision with a drone will do no harm, why the fuss? It is not at all surprising that the risk of drones to the safety of aviation is being worried about by members of the industry, with most respondents declaring that more regulation is required soonest. This is countered by a general feeling of regulatory change-fatigue!

It can be seen from my research that human error is inevitability in the aviation industry and that the way that it is managed is key to preventing accidents. This management often consists of a rationalisation of previous near misses and errors into a single cause, which is often blamed on human factors, but if further investigated could be seen to be rooted much deeper in equipment design or in regulation. Encouragingly, much of the industry is felt to have a just culture, with the perception of risk being communicated between managers and their staff appropriately. Whilst there are plenty of proponents of a Safety Culture, there are many differences of opinion as to what constitutes one. What constitutes error is well understood, how best to manage it in relation to risk is not, especially during any regulatory change.



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"Flying To Europe for the First Time"



By First Officer Faisal Hayat – B777 PIA

Many pilots these days are transitioning to Boeing 777 flying to new airspaces like Europe.

First order of business for a good preparation for a flight starts with getting an old flight plan and setting it up on your iPad.

Now that we know, the FIRs we will be flying, we can read about the differences they have filed in the Jeppesen.

4) Most of the time name of the Major city is the name of the FIR. This major city is also most often the best choice to land, if you have to land as soon as possible. For example, "Aktobe", "Samara", "Moscow" are great cities enroute to Europe from Asia, with significant airports.

Perhaps, we can download an ATLAS on our IPAD. Some non-aviation history and information about the countries we fly over can make the flight more interesting.

"Briefing notes" on our IPAD jeppesen, are a great help for planning our arrivals and departures from new airports that we are flying too. I feel, we must always meticulously read them, every time we fly to these airports, regardless of our experience.

The "Briefing notes", include information about preferential runway upon our arrival. Noise abatement procedures, taxi routes, Data link procedures, notes on using the APU, low visibility procedures, "De-icing" pods and procedures, lighting systems at the airport. The briefing notes are a goldmine.

Wikipedia also provides informal background information about all airports. This information can often put things into perspective.

Use that "phones a friend" option and life-line. Your friend can tell you gate number usually used by your airline.

STARs and SIDs are named usually as the names of our last and first waypoint on our flight plan.



By Nop Briex



By Sameer Haqqi

All information and preparation is of course always tentative. But the preparation of expected also prepares us for the unknown that we may face.

Always and always we must write down frequencies and clearances as we receive them. Compared to our regional Middle Eastern flight environment, the European airspace can be more demanding in this regard.



Look. LIVES ARE AT STAKE! Listen FOCUS

- **IT CAN HAPPEN TO YOU:** You may accidentally takeoff without a clearance by confusing an ATC instruction such as "Call when ready for departure" or "Roger" to mean "Cleared for takeoff."
- **THE FIX:** Write down ATC clearances or make a mental note to only act on a departure when you hear "Cleared for takeoff."



Federal Aviation
Administration



FAA Safety Briefing

 **PAKISTAN**
International Airlines
Great People to Fly With

Fire at Regent Plaza (Karachi)



By Naseem Sehri – Instructor, PIA Training Centre, Karachi.

This is the story involving reportedly 19 fatalities, 75 injuries & PIA Cabin crew's unsung bravery. On the fateful night of 05th December and wee hours of 6th December 2016, PIA Cabin Crew members from three outstation bases Islamabad, Lahore & Peshawar (ISB, LHE, PEW) were in deep slumber occupying rooms on the 5th, 6th & 7th floor of the Regent Plaza Hotel, Karachi.

PEW based crew had operated a flight from Kuala Lumpur, arrived at the hotel, had dinner & had gone to sleep.

ISB based crew had come to Karachi (KHI) for their annual Safety & Emergency Procedures (SEP) Training at PIA TRG Centre (PTC), but unfortunately they had been offloaded from two flights for ISB due to non-availability of seats (They were at the airport since the morning of 05th Dec' & were exhausted by the time they got to bed & again they were to wake up by 0600 on the 06th Dec. Some excerpts from conversations with various crew members PIA (though all cannot be listed here) are given below;

:: FS Imran Javaid | PEW Base ::

As luck would have it, in the early hours of 06th Dec at around 03:00, I woke up to the sound of smashing glass..... I jumped out of bed, pulled the curtain aside & saw huge flames of fire rising from the restaurant side! Instead of panicking, I swiftly changed from my night suit, collected my watch, joggers, torch, cell phone PIA ID Card & a towel.

The hotel was filled with thick dense smoke (You must have heard smoke kills faster than fire!).

I remained calm while I prepared to deal with disaster.... I reminded myself "I am not an ordinary average citizen.....I am a trained crew member expected to be tested in times like these!"

I knew the fire evacuation period for a building is about three minutes & if would be able to do just that, then I better be calm & in control of myself and for that I said to myself the following things to on priority basis;

- ✓ Panic Control
- ✓ Judgment
- ✓ Co ordination

I then started off by telling other guests & crew members to remain calm & started knocking the rooms of other guests / crew to warn & enable them to get out as soon as possible. I was in room 517 - 5th floor, I assisted some ladies from PIA crew to the end of the corridor to a sign that read upon opening it, we discovered it was a storeroom, quickly we turned to another large door next to the store room.....but to our horror a lock displayed the hopelessness of it all.

Though I had once suffered from spinal medical issues and have also been forced to work on ground for a long time due to this problem, but at this moment when I could not see much options I started kicking the door with all energy and strength that I had, I kept kicking it till it opened, letting in unbearable smell & thick black smoke from the other side.



With my torch, I was able to see the windows down the Emergency Exit stairs. As I started to break the windows with my torch & bare hands I kept shouting to others to bend low, cover their nose & mouth.

One guest caught my hand & asked me to accompany them, but I refused because my colleagues were apparently stuck on the 6th & 7th floor.

Huma Kalsoom & Sehrish responded to my calls & came out of their rooms. They didn't even know fire had mangled the hotel.

It is worth mentioning that these two brave ladies not only followed my instructions to go get towels & shawls, infact they started to assist me & others in evacuating as many guests as possible. We sent the next batch of people making them move swiftly, bent low. (Most people did not know the location of this Exit & were searching for the elevator, when directed by me to hurry up & leave;

Flight Purser (FP) Naeem Abbas refused to go saying "I am with a wheel chair guest & cannot leave without helping him to safety."

Suddenly I remembered my friend Flight Steward (FS) Kamran was on the 7th floor (Remember smoke is lighter & tends to rise upwards faster7th floor must be the thickest smoke, I thought & subconsciously started to pray for his wellbeing.... I will help the 7th floor guests to the open roof top, till rescue arrives I thought.... yet again another challenge awaited us.... **there was no apparent door to the rooftop!**

I was exhausted & started to suffer from severe smoke inhalation.....& yet I could not bear the thought of leaving my friends behind, so I kept chanting their names till FS Kamran & FP Khan Aslam responded with Imran Bhaae ap kahan hein (Where are you Brother Imran)?

I could see PIA Crew leading some other guests, shouting instructions, & hugging each other while running downstairs. By now smoke was so dense one could touch it....holding the staircase railing I started to slide down because this was my last chance. Along came 30 more people. We luckily evacuated but to my amazement, no ambulance came till now. We moved away to a safe distance.

I observed some crew members were missing....since I was the only one now outside the hotel with a cell phone, I started to call & direct them to that part of the hotel less affected by the fire.

Out came FP Naeem Abbas, FP Khan Aslam & HumaKalsoom. FS Kamran's foot was bleeding profusely with a shard of deeply embedded glass & so was my hand injured. I called Khi & Pew Crew Briefing Rooms and soon after, PIA Ambulance reported promptly. We sent the girls to the hospital first. Guests without cell phones were offered to make use of our phones. We tried to reassure some who were terrified & had to leave friends behind. When rescue services arrived called it a day & headed to the Agha Khan Hospital and eventually I received six stitches to my hand.

I tried to live up to my belief of living for others as well.

I feel dealing with Fire, terrorism & natural disaster should be taught in schools like other countries. Also First Aid should be made mandatory considering the volatile situation of our country. Prevention indeed is better than cure! Even countries like Africa have First Aid training in schools. This country most affected with accidents, terrorism & natural disasters should put it on war footing!!

ALLAH willed us to be alive & Shukar Alhumdullilah saved us.....After Allah we give credit to the PIA Training Centre for having hammered into us a preparation to deal with something like this which came without warning in a place most of us like to believe as safe!

DGM Training Ms. Ruxana Akmut spent the whole day looking after us. Chief Instructor Flight Services (CIFS) Ms. Kaleem sent one Instructor who stayed & assisted us through the afternoon.

I need to say this because we weren't left unattended by our respective departments, as is the case we generally hear in the news of other organization employees country wide. Flight Service Dept. & Flight Service Unit / PTC need to be credited for a quick and efficient response for all of us and making us feel secure.'

:: Senior Purser Razia | ISB Base ::

'I soaked a pillow when I heard screams and thick smoke coming from under my door I then soaked the blanket and covered myself as I ran out, yet smoke kept penetrating through.... the room next to me had its door open & i saw two women lying dead on the floor.....it was traumatic!

People were dead even before we had evacuated! Allah helped us survive this tragic event, the prayers of our well-wishers & PTC who helped us train & refresh each year for incidents like this.

I think crew members undergoing an incident like this should have formal in-house post trauma counseling. Surprising is the fact that the hotel had no serviceable smoke detectors to warn occupants of danger.

After being released from AKU our training was postponed & we returned to PTC after one week.

When at PTC, Principal PTC Mr. Sohail Mahmood, CIFS and all Instructors & trainees gathered hugged us & gave us flowers on being blessed with life for the second time!'





During the sad event, Airhostess (Ah) Ramla Nawaz, from her room on the 4th floor, finding no way to escape tied two bed sheets with a knot and tried to shimmy down. While doing this she lost her grip on the bed sheet & fell to the second floor on shattered glass, breaking her leg at two locations.

Ramla, a brave young lady is undergoing extensive treatment for recovery and We wish her a speedy recovery. 27 crew members were taken to AKU on 06 Dec 2016. A date on which they experienced two very strong emotions... horror & then gratitude. Some were released from the hospital by the afternoon after initial treatment while others had to stay longer. G.M FSD Naeem Yaqoob reached the scene and was seen supervising the treatment personally for his crew members till most were released by the afternoon. Emergency Response Team of PIA also reported to AKU took details & assisted crew to return to the hotel.

PIA is indeed proud of her crew members' exceptional conduct of courage & responsibility when called upon in an emergency challenging their response as trained emergency personnel. We not only teach them technicalities but also expand empathy.

Take Home Lessons;

- ✓ Always know the number for fire Department
Nationwide dialling: 16
- ✓ Always know the emergency exits
- ✓ Never use elevators in case of Fire
- ✓ Train and be prepared for contingencies

Symbols Call for Attention



In emergency, Assembly Area



Fire Extinguisher here



Caution Symbol



Exit in Emergency only



Emergency Stop Button



Activate Fire Alarm

“QUIT SMOKING”

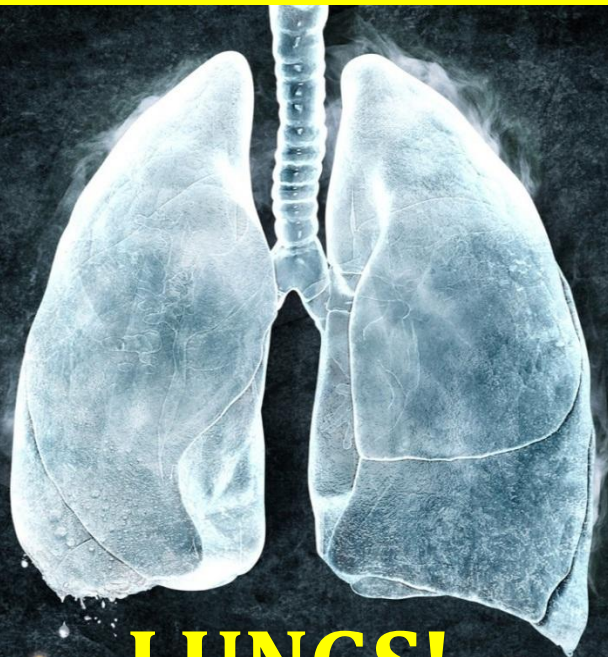
It's Injurious to your health!

You are done listening to it,

...and we are done telling it!

Just One Last Thing...

**STOP
BURNING
YOURS &
OTHER'S**



LUNGS!

PIA Management is fully committed to provide smoke free environment.

:: PIA Policy Statement ::

“Smoking shall be prohibited within ten (10) feet of PIA building entrances and exits.”



Captain Naveed A. Aziz



Safety Chief Captain Naveed A. Aziz was engaged with Pakistan International Airlines on 21st Feb 1977. Presently holding the position of Chief Safety Officer (CSO) as well as performing flying duties as a Captain onboard Boeing 777. He shall be superannuating on 30th July 2017 after serving the airline for more than 40 years.



Memories may be beautiful and yet what's too painful to remember we simply choose to forget. So it's the laughter we'll remember when we remember the way we were.

Assignments on Executive Level:

Chief Pilot Standard Inspection	- 1 st Dec 2006 to 24 th May 2007
Chief Pilot Technical Operations	- 25 th May 2007 to 30 th Mar 2008
Director Flight Operations	- 31 st Mar 2008 to 09 th May 2008
Director Flight Standards, CAA	- 18 th Apr 2009 to 27 th June 2011
Director Flight Operations	- 28 th Jun 2011 to 28 th May 2012
Director Safety Management	- 16 th Oct 2015 to 14 th Dec 2015
Director Flight Operations	- 15 th Dec 2015 to 22 nd Feb 2016
Director Safety & QA/CSO	- 25 th Feb 2016 to Present



Career Progression as a Captain:

Cadet Pilot	:	21 st Feb 1977
First Officer F27	:	16 th Jun 1977
First Officer B707	:	2 nd Jan 1979
Captain F27	:	21 st Apr 1983
First Officer A300	:	31 st Jul 1986
Captain B73	:	17 th Oct 1991
Captain A310	:	3 rd Apr 1996
Captain A300	:	30 th May 2000
Captain B777	:	13 th Apr 2004

GOODBYE



True friends don't say good-bye; they just take extended leaves of absences from each other.

FLIGHT DATA MONITORING

Part-III



By Engr. Syed Ahmed Kamran – DCE FDM

:: Flight data processing and analysis for FDM ::

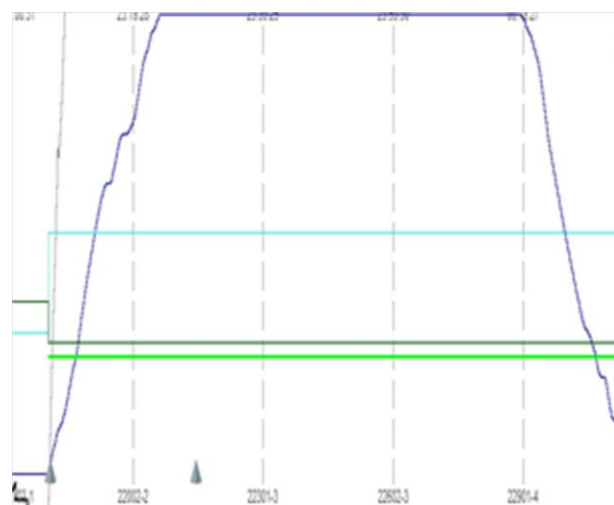
The flight data received by the FDM lab from the aircraft recorder is normally a single file from the SSFDR or a group of files from the QAR or FDMU/FDAU. The format and size of the files can differ depending on the manufacturer and configuration of the recorder. File types such as *.FDR, *.DLU, *.QAR are common.

The flight data in these files called raw data files, is encoded as ones and zeros and is not understandable by just looking at the data as can be seen from the figure below.

Frame-Sf	SfCount	Sf	Word	U1	U2	U3
21703-1	86812		00	000001000000	000011110000	000000000011
			09	011100111000	011110011011	000000010000
			12	001111111101	111111010011	111111010100
			1B	111110110110	111111010000	000101111110
			24	000110010000	100001001111	100010010100
			2D	000000000000	111111111111	111111110000
			36	000000000000	000000000000	000011010111
			3F	001110110111	101001110010	101001111000
			48	111111111011	111111110111	000000000010
			51	111111101010	111111110101	111111110101
			5A	000100111111	000101110111	000010010000
			63	001000101001	001000100101	001000101110
			6C	111111100000	000000000110	000000000000
			75	001100000001	001100000111	000101110000
			7E	000001010101	100100000000	000000000000
			87	000010010011	000000000000	000000000001
			90	000010100000	000011000011	000010000000
			99	000000000010	000000000101	000000000100
			A2	010000110101	010000111000	010000111000
			AB	000000000110	000000001101	111011001100
			B4	000000000000	000000000000	000000000000
			BD	000000100101	000000100011	000000100011
			C6	000010110000	111111010101	000000000000
			CF	000001101110	000011101111	000001101001
			DE	000000000000	011001000001	000000000000

To understand this data it has to be converted to engineering values and be presented in a tabular or graphical format as shown below.

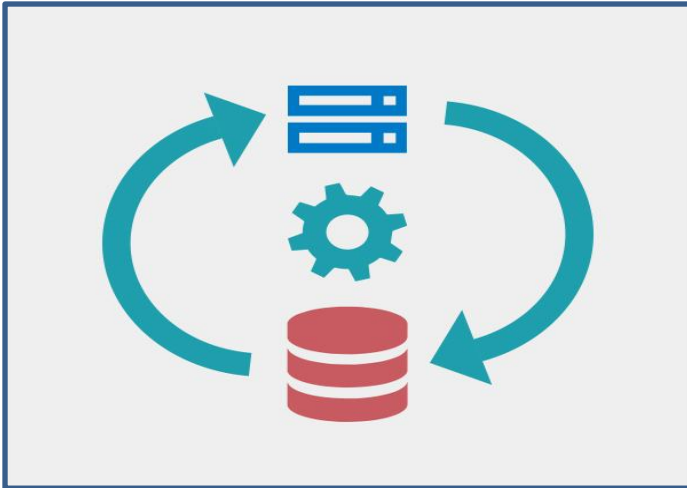
Frame-Sf	SfCount	Time St	RADIO HEIGHT aRALT FEET	STANDARD ALTITUDE ALT STD FEET	LANDING GEAR STATUS _LDG_STATUS	GEAR SELECTION UP _LDG_SEL	LDG GEAR CTRL LVR DN aLDGSEL
21703-1	86812	22:00:31	-9	1949	FULL GRD	DOWN	DOW
21703-2	86813	22:00:32	-9	1949	FULL GRD	DOWN	DOW
21703-3	86814	22:00:33	-9	1949	FULL GRD	DOWN	DOW
21703-4	86815	22:00:34	-9	1948	FULL GRD	DOWN	DOW
21704-1	86816	22:00:35	-9	1948	FULL GRD	DOWN	DOW
21704-2	86817	22:00:36	-9	1948	FULL GRD	DOWN	DOW
21704-3	86818	22:00:37	-9	1949	FULL GRD	DOWN	DOW
21704-4	86819	22:00:38	-9	1949	FULL GRD	DOWN	DOW
21705-1	86820	22:00:39	-9	1948	FULL GRD	DOWN	DOW
21705-2	86821	22:00:40	-9	1948	FULL GRD	DOWN	DOW
21705-3	86822	22:00:41	-9	1948	FULL GRD	DOWN	DOW
21705-4	86823	22:00:42	-9	1949	FULL GRD	DOWN	DOW



Processing of raw data files: The processes of converting the raw data files from zeros and ones to engineering values requires a software application tool that decodes the file. However the software tool in itself is not capable of reading any raw data file unless it has been programmed with the conversion matrix for the specific file format. This conversion matrix is available in a document, known as the “data frame layout” produced by the FDAU/FDIMU manufacturer. The DFL is the most important document that every airline must have to develop and maintain an accurate data processing system. Normally one DFL is good for one aircraft type if all of the aircrafts are equipped with the same FDMU/FDAU. Having too many different FDMU/FDAU for the fleet not only increases the initial cost of programming the conversion matrix, but also costs more to maintain.

As it was mentioned in part 1 of this article published in the earlier safety magazine issue, that PIA has purchased the AirFASE application software tool to process and analyze the flight data. This tool is now also available with other regional airlines. Even though AirFASE is not the only tool available in the market, but it definitely has a very big customer base. Teledyne controls is the developer of AirFASE and is continuously upgrading the product to make it better.

Once the raw data is processed, all recorded parameter values can now easily be read for the entire duration of the flight.



Depending on the size of the raw data file, there can be as many as 300 flights data recorded. For purposes of investigation into incidents, the tabular and graphical format is more than sufficient. However, for purpose of FDM, a little more elaborate display is required to perform the analysis.

Analysis of Events: The purpose of the FDM program is to determine if the flight operations was performed within the defined standard operating procedures. Now to look into the tabular data and check for the deviations would be a very tedious task, especially if we had to look at thousands of hours of data containing hundreds of parameters. For this purpose, the AirFASE software has, what is called a flight analysis program (FAP). A single FAP is specifically written for each aircraft type. In the FAP a few hundred events are defined that are actually small codes with trigger conditions that look out for deviations in parameter values through-out the flight. Once the program code finds a deviation it automatically highlights it as an event and displays it on the screen. Several events thus displayed are then available for analysis by the analyst. A sample display on AirFASE is as below.

Results											
Flights											
AC Type	Evt	Sev.	Flight #	Depart. (GMT)	Duration	Origin	Dest.	Reg. #	Fit Upload	Comments	
ATR42	4	Low	682	24/05/2017 16:26:49	01:31:27	MUX	ISB	AP-BHM	25/05/2017 11:31:41		
ATR42	2	Low	681	24/05/2017 14:36:33	01:07:07	LHE	MUX	AP-BHM	25/05/2017 11:31:41		
ATR42	3	Low	681	24/05/2017 12:37:32	00:59:32	MUX	LHE	AP-BHM	25/05/2017 11:31:41		
ATR42	6	Medium	681	24/05/2017 10:22:30	01:34:59	ISB	MUX	AP-BHM	25/05/2017 11:31:42		
ATR42	6	Medium	589	24/05/2017 08:37:48	01:37:27	DEA	KHI	AP-BHH	25/05/2017 11:22:04		
ATR42	12	High	589	24/05/2017 06:19:45	01:39:39	KHI	DEA	AP-BHH	25/05/2017 11:22:01		
ATR42	7	High	606	24/05/2017 03:43:50	01:00:08	GIL	ISB	AP-BHM	25/05/2017 11:31:42		
ATR42	17	High	605	24/05/2017 02:00:20	00:52:28	ISB	GIL	AP-BHM	25/05/2017 11:31:42		
ATR42	4	High	682	23/05/2017 18:18:06	01:00:15	GIL	ISB	AP-BHM	25/05/2017 11:31:44		
ATR42	2	High	682	23/05/2017 16:35:27	00:59:59	ISB	GIL	AP-BHM	25/05/2017 11:31:44		



Now there is a lot of information in the above display, and it is the job of the analyst to determine the validity and seriousness of the event for ensuring safety of flight operations. Obviously there are two things to note, one is that the data the analyst is looking at is of a past flight and secondly none of it is actually an immediate cause for concern. However, it is an indication of an unsafe trend developing within the airlines flight operations that needs to be monitored, checked, verified and mitigated to make flight operations safe. In part IV, we will cover the remaining aspects of flight data monitoring, such as data reporting, the FDM team and Flight Analysis Program development.

Conflict Resolution



By First Office Jehanzeb Khan – DGM SMS

It is of first concern to us that why should a conflict rise in the first place between two or more than two parties involved in an activity together or having common interest?

Another question; is conflict as bad as we may consider it to be after reading the first sentence of this article? Is collaborating together joining hands and minds, compromising on situations and consensus always a good alternative to having different approach and opposition? Importantly, we shall first allay the effect of the word conflict in order for our discussion to be a little more meaningful, a conflict does not necessarily mean friction, bad intentions or tug of war. It is actually the terminology that may sound confusing as it is a word taken from common English, otherwise a difference of opinion is what it actually is and being part of professional environment and working as professionals we shall all understand that emotions shall be kept out of our professional work, now again it does not mean that we shall not have empathy for the other colleagues or sense of ownership for the organization we are working for, but instead we shall think more rationally and weigh our decisions and try not to compete for ourselves and individual decisions but instead for the larger good, the better results for the company and its stake holders within the envelope of safety and efficiency. Difference of opinions is a healthy sign of thinking minds and shall not be disregarded or disliked at any stage for us to improve, build better platforms and expand productively but, undue friction is not of any good to either individuals or the team, so decisions shall not be held at abeyance for unnecessary brain waves that may be produced for irrational and ostensible reasons. To simplify, we shall have our own ideas and these may be communicated if it's well thought and has background of logic and thought and not mere hunch or vibes.

As practically experienced and studied in our own working atmosphere, undue conflicts along with feeling of discontent arise from weak relationship between the two parties, be it the two pilots flying a flight machine, two individuals of the same department or the two departments/divisions they themselves.

This not being the only reason but, is the major factor for disgruntled work life. We not realizing that this along with limiting our working potentials as teams at our workplaces, also affects our health and overall wellbeing; A study about 'Adult Development', underway at Harvard University which is in place since last about 75 years, involving 724 men was utilized to find 'what makes people happy' and whom are the content most, we shall not be surprised to hear that the ones having good relationships in their lives (work and home) were the happiest most and nothing including financial status or fame could replace it. So we shall look for a Win-Win situation for the jobs and ourselves by enhancing our individual skills to have a better bonding amongst us and our colleagues.

The second factor that constitutes being the major causal factor within our organization's occasional undue friction is 'structural conflicts' that arise due to the factors outside the domain of the parties involved and may be due to limited physical resources, limitation of practical authority despite it been penned down, frequent hierarchy changes in the management, time constraints and also physical distances of the parties etc.

One of the phenomena that 'structural conflict' induces is the feeling and negative thoughts of considering your own team or individual self as the unit that is working fine but, the other units (individuals, divisions or departments etc) are considered redundant or ineffective and inefficient and hence this causes good teams to sunder into practically ineffective fragments that may be working harder than they have to if they were united as one team but still producing insufficient results.



So, an antidote to this would be to evolve our thinking and understand that just like us the individual on the other side of the table is also restricted and facing issues and hence the possibility of the best results not yielding can be due to this and not the incompetence.

The above issue has also apparently caused top boss's anxiety and confidence at the same time (be it the office head or an airliner's commander), confidence because they do foresee the possibility of achieving goals keeping their experience and team capabilities in mind and anxiety due to undue fear of incapability of achieving goals in time due involvement of all these external influences and other factors. One of many antidotes for this lies in Boss's requirement to gain his/her worker's trust by being empathic, helpful, man of his words and being goal and relationship oriented at the same time.

Now let's discuss in short, the strategies or styles of conflict management as given by Thomas and Kilman; Any conflict can be managed in one of the five styles, namely;

1. Avoiding
2. Competing or Dominating
3. Obliging
4. Compromising
5. Integrating

As seen from the picture, one can incline towards being relationship oriented; task oriented or even can be witty enough to achieve both at the same time.

The ones that concern me the most are the "Avoiding" styles in which a person becomes apathetic and shows signs of resignation, not caring for themselves or the other involved party.



Units of an organization having permanent "Obliging" approach may not be the best for being in the decision making positions, the reason for that is evident from the very nature of these as they would opt for the way While the "Dominating" style apparently seems undesirable but is useful if used in some situations, as in this one party losses while the other takes home the trophy, so it does not always have to be considered negatively, instead a fair competition of ideas or efforts can be held and the winning party is the one whom can substantiate its opinion, with no hard feelings to either of the involved (some sort of sportsmanship I would say) in which the other party is happy while they lose.

The ones of the "Compromising" view will end up with accomplishing some of their objectives while would accept some of the other parties too, even if it is not in their own interest. These tactics are somewhat street-smart and eventually keep the ball rolling

The best possible approach but not always practically achievable is the Win-Win situation through integration of the strategies brought up by the involved or consenting parties.

We can do further reading through vastly available material on the internet for the model described above, but it shall be understood that no single style is always the best for all situations to manage clashes, instead it is the particular situation in the given time that requires either exactly one style or may be more than one can do the job.

Finally, it is to be understood by us as individuals and our team units that the whole mammoth organization comprises; difference of opinions or conflicts of ideas are productive phenomena but depends that what we intend by it and how it is handled or even utilized by the individual(s) managing the situation. So, lets keep it simple and not feel offended by someone saying a "NO" to you even for the best of your ideas, lets solve them and not take these personally because an individual not thinking rationally but emotionally will make more mistakes than the one who thinks logically. ▲

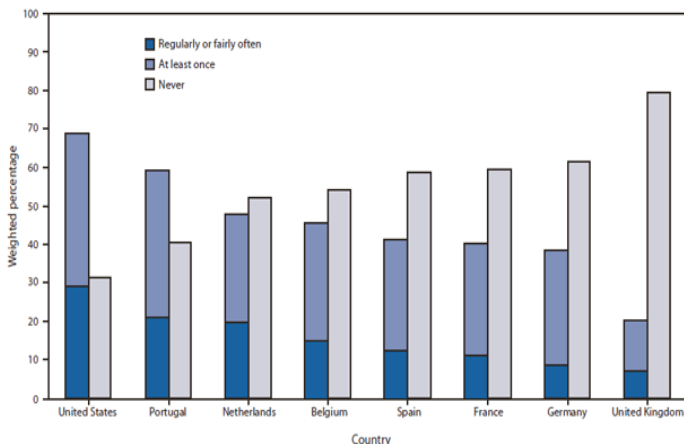




A prescription for Disaster!

By Latafat Mehmood | Manager HSE

With cellular phone popularity reaching new heights (hundreds of millions of cell phones are in use in Pakistan), more and more drivers are using their cell phones to conduct business and personal affairs. It is not surprising that people will attempt to optimize their time in the vehicle by using phone and computers. On the surface it looks like a great way to make use of the time, but there is a dark side: Dialing, discussing and doing deals all affect your ability to properly respond to typical road hazards -- let alone challenging driving situations. Whenever you're driving a vehicle and your attention is not on the road, you're putting yourself, your passengers, other vehicles, and pedestrians in danger. Stressful or heated conversations, your passengers, other vehicles, and pedestrians in danger. Stressful or heated conversations, especially those involving relationships, contribute to driver distraction.



When combined with a cell-phone call, the combination can be deadly. There's also a difference between driving while talking on the phone and driving while chatting with a passenger. Passengers in the car often alert drivers to dangerous situations, in sharp contrast to cell phone callers who are oblivious to a driver's surroundings. Reports and surveys show that mostly drivers are observed to be distracted while talking on cell phones as they drift into other lanes or run through red lights or stop signs. In some cases, the results have been even fatal.

"Avoid using cell phones while driving. This is disastrous!"

**BETTER LEFT UNREAD
THAN DEAD.**



WATCH YOUR STEPS



1. Keep cord and wires out of way.
2. Keep your work area clean.
3. Make sure that carpet and rugs are free from holes and loose edges.
4. Look out for area with poor lighting and report it.
5. Be alert, never get hurt! 😊



Think About Caption



Suggest a caption for this photo;

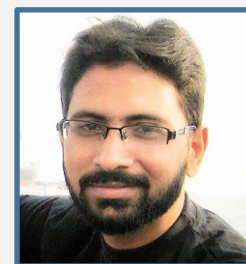


Send your caption on pkhse@piac.aero along with your photo, name, P.No & designation. Best caption selected by magazine editorial team, with details of sender will be published in next magazine issue. We welcome all of you, for your best captions!
-- Editorial Team

"Seconds from disaster..!"



This photo was published in PIA Safety Magazine Issue-3 (Feb 2017), and this best caption selected for this photo was sent by;



Muhammad Mehmood Iqbal
Senior Technician | P-66417
PIA Engineering & Maintenance

“Aviation Quotes”

“Aviation in itself is not inherently dangerous. But to an even greater degree than the sea, it is terribly unforgiving of any carelessness, incapacity or neglect.”

Captain A. G. Lamplugh

“Aviation is least forgiving of mistakes.”

Freeman Dyson

“A good landing is one from which you can walk away, and a great landing is one after which they can use aeroplane again!”

Captain A. G. Lamplugh

“In flying I have learnt that carelessness and overconfidence are usually more dangerous than deliberately accepted risks.”

Wilbur Wright

“Aviation is proof that given, the Will, we have the capacity to achieve impossible.”

Captain A. G. Lamplugh

“Excellence is not a skill, it's an attitude.”

Ralph Marston

“Never quit. Never give up. Fly it to the end.”

Chuck Aaron

*“In the sea there are countless gains,
but if thou desirest safety, it will be on the shore.”*



Safety & QA Department

THINK

THE
BEST
SAFETY
DEVICE
IS A



CAREFUL WORKER!

Get the Safety Habit!

PIA AP-BMG B777-2Q8ER

Photo by Abdul Ahad



PIA AP-BID B777-340ER

Photo by Saheed Zaman

